

Magnetic reconnection

12.1 Introduction

Section 2.6.4 established the fundamental concept underlying ideal MHD, namely that magnetic flux is frozen into the plasma. This concept means that the magnetic topology of an ideal MHD plasma cannot change because a change in magnetic topology would require a change of magnetic flux within the frame of the plasma. Chapter 10 showed that ideal MHD plasmas are susceptible to two distinct types of instabilities, pressure-driven and current-driven. Pressure-driven modes draw on free energy associated with heavy fluids stacked on top of light fluids in an effective gravitational field whereas current-driven instabilities draw on free magnetic energy and involve the plasma attempting to increase its inductance in a flux-conserving manner. Both of these instabilities occur on the Alfvén time scale defined as some characteristic distance divided by v_A .

It is possible for an MHD equilibrium to be stable to all ideal MHD modes and yet not be in a lowest energy state. Because ideal MHD does not allow the topology to change, a plasma that is not initially in the lowest energy state will not be able to access this lowest energy state if the lowest energy state is topologically different from the initial state. However, the lowest energy state could be accessed by non-ideal modes, i.e., modes that violate the frozen-in flux condition, and so the available free energy could drive an instability involving these non-ideal modes. Magnetic reconnection is a non-ideal instability where the plasma is effectively ideal everywhere *except at a very thin boundary layer* where the ideal MHD frozen-in assumption fails so magnetic fields can leak across the plasma and change their topology. Even though this boundary layer is microscopically thin, the reconnection and associated change in magnetic topology at the boundary layer allow the configuration to relax to a lower energy state. Magnetic reconnection thus describes how a very slight departure from ideal MHD leads to important new behavior.

The simplest reconnection model is obtained by including finite resistivity in the MHD description, but more elaborate non-ideal physics is also possible due to two-fluid or Vlasov effects omitted from MHD (e.g., finite electron mass, Landau damping). Resistive reconnection is much slower than ideal MHD modes because it involves the diffusion (i.e., leakage) of magnetic field across the plasma in some very limited spatial region where ideal MHD breaks down. Although including finite resistivity in an MHD description is the simplest way to invoke non-ideal physics enabling magnetic diffusion, the actual resistivity of most relevant laboratory and space plasmas is inadequate by orders of magnitude for observed reconnection rates to be explained by resistive MHD. Observations and computer models show that the observed rapid reconnection rates are due to complicated two-fluid and Vlasov effects, typically involving wave excitation and acceleration of particles to very high energies.

Nevertheless, analysis of resistive MHD provides a good introductory overview of the main issues involved in magnetic reconnection. In fact, even though it is known that two-fluid and Vlasov models would be more appropriate, resistive MHD is often used as a convenient approximate description for actual situations by invoking a suitably large ad hoc “anomalous” resistivity. This is a hybrid approach that essentially separates the reconnection physics (e.g., location, rate, topology) from the question of what causes the dissipation enabling the reconnection. The anomalous resistivity approach simply assumes there exists some complicated collisionless mechanism (e.g., some form of wave turbulence) for slowing down electrons relative to ions and leaves the issue of determining this mechanism as a separate problem. Investigation of collisionless reconnection is an active research area at the time of writing and a single straightforward theory does not exist. Resistive MHD analysis should therefore be considered as a first step in attacking the problem and a means for identifying relevant issues, but not as a quantitatively accurate model for most real situations where complicated collisionless physics seems to be much more important than the effect of finite resistivity.

Magnetic reconnection (also known as tearing) is an intrinsically complicated boundary layer phenomenon because two very different length scales mutually interact. This imposes a strong burden on the mathematical description because the mathematics has to simultaneously characterize two different kinds of physics, namely (i) the diffusive physics of the non-ideal, microscopically thin reconnection layer and (ii) the ideal behavior of the macroscopic regions on either side of this thin layer. We shall begin the discussion with an analogy drawn from everyday experience and then use this analogy to argue why it is energetically favorable for a sheet current to break up into filaments. The time scale for the resistive MHD reconnection process will be estimated and shown to be much slower than ideal MHD physics. It will then be shown that the existence of current sheets

corresponds to having β of order unity and so the current sheet model is not directly relevant to low- β plasmas. However, it will be shown that a modest generalization of the current sheet analysis to the situation of sheared magnetic fields provides a model for low- β situations.

12.2 Water-beading: an analogy to magnetic reconnection

Since magnetic reconnection is not a simple process, it is helpful to start by considering the dynamics of a somewhat analogous instability known from everyday experience, namely the process of water beading. The initial condition in water beading is shown in Fig. 12.1(a) and consists of a long, thin, two-dimensional, incompressible drop of water (approximate analog to the magnetic field) frictionally attached to a substrate (approximate analog to the plasma). The long, thin drop has surface tension “trying” to reduce the perimeter of the drop (this is analogous to the pinch force being interpreted as field line “tension” squeezing on a current channel). If the drop were not attached to the substrate then, as shown in Fig. 12.1(b), the surface tension would simply collapse the long, thin incompressible drop into a circular drop having an area equal to that of the initial long thin drop. Because of the frictional work involved in dragging the water over the substrate, this wholesale collapse is not energetically favorable and does not occur. On the other hand, if as shown in Fig. 12.1(c) the long, thin drop breaks up into a line of discrete segments (which does not involve significant dragging of

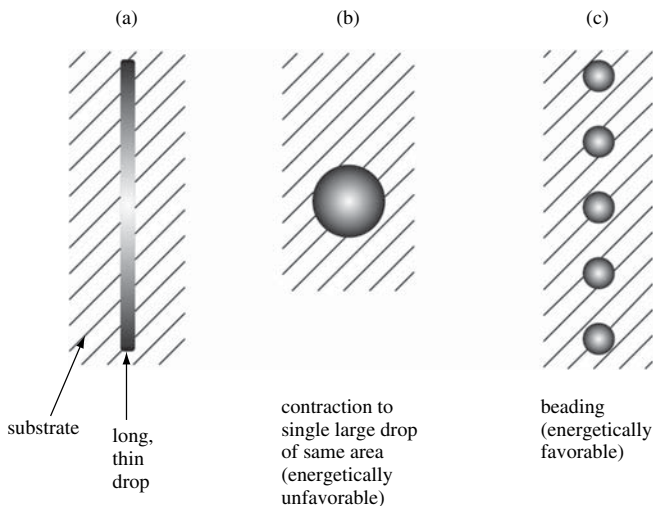


Fig. 12.1 (a) Initial long, thin drop, (b) contraction into big, round drop (too much scraping on substrate), (c) beading into chain of little drops.

water across the substrate), the surface tension of each line segment causes each discrete segment to contract in length and bulge in width until circular. Because only modest frictional dragging of water across the substrate is required to do this, the process can be energetically favorable, and so result in water beading.

12.3 Qualitative description of sheet current instability

Suppose an infinite-extent plasma has, as shown in Fig. 12.2(a), a thin sheet of current flowing in the z direction (out of the page). The sheet current is centered horizontally at $x = 0$ and extends vertically from $y = -\infty$ to $y = +\infty$. The current is uniform in both the y and the z directions and roughly corresponds to the initial long, thin water drop. This somewhat artificial situation can be considered as the Cartesian analog of a cylindrical shell current flowing in the z direction, localized at some radius $r = r_0$ and azimuthally symmetric. Thus, we identify r with x and y with θ .

Since all quantities depend on x only, Ampère's law reduces to

$$\mu_0 J_z(x) = \frac{\partial B_y}{\partial x}, \quad (12.1)$$

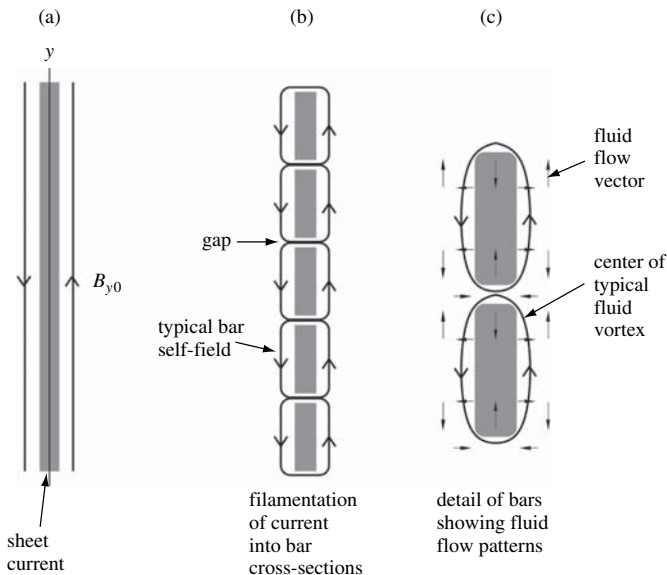


Fig. 12.2 (a) Initial sheet current (x axis is in horizontal direction, y axis is vertical, z axis is out of page); (b) breaking up into bar cross-section filaments; (c) detail showing fluid flow consists of set of small vortices that are antisymmetric in x .

which integrates to give

$$B_y(x) = B_y(0) + \int_0^x \mu_0 J_z(x') dx'. \quad (12.2)$$

On the basis of symmetry the sheet current cannot generate a magnetic field at $x = 0$ so the field $B_y(0)$ must be entirely due to external currents. Let us assume for now that no external currents exist and set $B_y(0) = 0$; the situation of finite $B_y(0)$ will be treated later in Section 12.5. Thus $B_y(x)$ is a sheared magnetic field that is positive for $x > 0$ and negative for $x < 0$. The magnitude of $B_y(x)$ changes rapidly inside the current layer and becomes constant for $|x| \rightarrow \infty$.

This situation can be characterized analytically by using a magnetic field

$$B_y(x) = B \tanh(x/L), \quad (12.3)$$

where L is the scale width of the current layer. Substituting in Eq. (12.1) gives

$$J_z(x) = \frac{B}{\mu_0 L} \cosh^{-2}(x/L), \quad (12.4)$$

which is sharply peaked in the neighborhood of $x = 0$.

Suppose, as shown in Fig. 12.2(b), a perturbation is introduced whereby the current sheet cross-section breaks up into a number of bar-shaped structures each with current I_{bar} separated by small gaps in the y direction. These bars of current (i.e., filaments) are the analog of the water beads discussed in Section 12.2. The pinch force due to the self-magnetic field of each bar acts like an elastic band wrapped around the bar (analogous to the surface tension of a drop). This tension contracts the y dimension of the bar and, if the bar is incompressible, the x dimension will then have to grow, as shown in Fig. 12.2(c). As the bar deforms from a rectangle into a circle having the same area, its perimeter becomes shorter giving rise to a stronger field (and effective surface tension) since

$$\int_{perimeter} \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{bar} = const. \quad (12.5)$$

Hence, this deformation feeds upon itself and is unstable. Note that the inductance of the system increases as the current sheet breaks up into current filaments, consistent with the discussion on p. 315 showing that a plasma lowers its potential energy by increasing its inductance. Instability is thus possible in principle, because the potential energy of the configuration would be lowered if the filamentation were to occur.

Before the gaps between the bars form, all magnetic field lines are open and straight, stretching from $y = -\infty$ to $y = +\infty$, whereas after the gaps form, some field lines circle the current filaments and so are closed. In order for the configuration to transform itself from the initial state to the filamented state, some

magnetic field lines must move across plasma and thereby change the flux linked by plasma (this change in topology is approximately analogous to dragging the water drops across the substrate in the water-beading problem). This motion of field lines across the plasma is forbidden by ideal MHD because it would violate the frozen-in flux constraint of ideal MHD. Thus, even though this filamentation instability is energetically feasible, it is forbidden in an ideal plasma, i.e., in a plasma governed by ideal MHD.

The field lines have an “X” shape at the center of the gaps and the change in topology occurs at these X-points. The center of the bar is called an O-point since the field around this point has an O-shape. If some non-MHD mechanism exists that allows these X-points to develop, an instability involving this mechanism becomes possible because of the available free energy. Since finite resistivity allows magnetic field to diffuse across plasma, a resistive plasma ought to be susceptible to this instability and the instability should occur at locations where the magnetic field attachment to the plasma is weakest. The location of the weak spot can be deduced by examining Ohm’s law for a nearly ideal plasma (i.e., small but finite resistivity),

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}. \quad (12.6)$$

At most locations in the plasma, the two left-hand side terms are both much larger than the right-hand side resistive term and so plasma behavior is determined by these two left-hand terms balancing each other. This balancing causes the electric field in the plasma frame to be zero, which corresponds to the magnetic field being frozen into the plasma. However, if there exists a point, line, or plane of symmetry where either $\mathbf{U}$ or $\mathbf{B}$ vanishes, then in the vicinity of this special region, Eq. (12.6) reduces to

$$\mathbf{E} \approx \eta \mathbf{J}. \quad (12.7)$$

The curl of Eq. (12.7) gives

$$\frac{\partial \mathbf{B}}{\partial t} = \frac{\eta}{\mu_0} \nabla^2 \mathbf{B}, \quad (12.8)$$

a *diffusion equation* for the magnetic field. Thus, in this *special region* the magnetic field is not frozen to the plasma and diffuses across the plasma.

Let us now examine the fluid flow pattern associated with the bars sketched in Fig. 12.2(b) as these bars contract in the y direction and expand in the x direction. As shown in Fig. 12.2(c), each bar has y -directed velocities pointing from the gap towards the bar center and x -directed velocities pointing out from the center of the bar. To complete the incompressible flow there must also be oppositely directed x and y velocities just outside the bar with the net result that there is a set of small fluid vortices that are antisymmetric in x and, in addition, have a

y -dependence 90° out of phase with respect to the y -direction periodicity of the bars. In particular, there is an outward x -directed velocity at the y location of the O-points and an inward x -directed velocity at the y location of the X-points. The fluid motion thus consists of a spatially periodic set of vortices that are antisymmetric with respect to $x = 0$. Each bar has a pair of opposite vortices for positive x and a mirror image pair of vortices for negative x , so there are four vortices for each bar.

12.4 Semi-quantitative estimate of the tearing process

An exact, self-consistent description of tearing and reconnection is beyond the capability of standard analytic methods because of the multi-scale nature of this process. However, the essential features (geometry, critical parameters, growth rate) and a reasonable physical understanding can be deduced using a semi-quantitative analysis, which outlines the basic physics and determines the relevant orders of magnitude. The starting point for this analysis involves solving for the vector potential associated with the magnetic field in Eq. (12.3), obtaining

$$A_z(x) = - \int^x B_y(x') dx' = -BL \ln [\cosh(x/L)] + \text{const.} \quad (12.9)$$

The constant is chosen to give $A_z = 0$ at $x = 0$ so

$$A_z(x) = -BL \ln [\cosh(x/L)]. \quad (12.10)$$

For $|x| \ll L$, $\cosh(x/L) \simeq 1 + x^2/L^2$ while for $|x| \gg L$, $\cosh(x/L) \simeq \exp(|x|/L)/2$. Thus, the limiting forms of the vector potential are

$$\lim_{|x| \ll L} A_z(x) = -\frac{Bx^2}{L} \quad (12.11)$$

and

$$\lim_{|x| \gg L} A_z(x) = -BL \left(\frac{|x|}{L} - \ln 2 \right). \quad (12.12)$$

Near $x = 0$, A_z is an inverted parabola with a maximum value of zero, while far from $x = 0$, A_z is linear and becomes more negative with increasing displacement from $x = 0$. This behavior of the vector potential is consistent with the field being uniform far from $x = 0$, but reversing sign on going across $x = 0$. The behavior is also consistent with the relationship between the current density and the second derivative of the vector potential,

$$\mu_0 J_z = \frac{\partial B_y}{\partial x} = -\frac{\partial^2 A_z}{\partial x^2}. \quad (12.13)$$

Current density is therefore associated with curvature in $A_z(x)$ or, in a more extreme form, with a discontinuity in the first derivative of $A_z(x)$. Specifying the vector potential is sufficient to characterize the problem since the magnetic field and currents are the first and second derivatives of $A_z(x)$ respectively. This general idea can be extended to more complicated geometries if there is sufficient symmetry so that specification of an equilibrium flux profile uniquely gives both the equilibrium field and the current distribution.

The reconnection process is characterized by the MHD equation of motion

$$\rho \frac{d\mathbf{U}}{dt} = \mathbf{J} \times \mathbf{B} - \nabla P, \quad (12.14)$$

Faraday's law expressed as

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \quad (12.15)$$

Ampère's law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad (12.16)$$

and the resistive Ohm's law

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}. \quad (12.17)$$

The analysis involves relating the velocity vortices to the linearized Ohm's law, and in particular to its z component

$$E_{z1} + U_{x1} B_{y0} = \eta J_{z1}. \quad (12.18)$$

The sense of the vortices sketched in Fig. 12.2(c) indicate that the velocity perturbation is uniform in the z direction and U_{x1} is *antisymmetric* with respect to x . Also, since the motion consists of vortices, there is no net divergence of the fluid velocity and so it is reasonable and appropriate to stipulate that the flow is *incompressible* with $\nabla \cdot \mathbf{U} = 0$. Since the perturbed current density is in the z direction, and since for straight geometries the vector potential is parallel to the current density, the perturbed vector potential may also be assumed to be in the z direction. Hence, both equilibrium and perturbed vector potentials are in the z direction and so the total magnetic field is related to the total vector potential by

$$\mathbf{B} = \nabla \times A_z \hat{z} = \nabla A_z \times \hat{z}. \quad (12.19)$$

Equation (12.18) can be recast using Eqs. (12.15) and (12.19) as an *induction equation*,

$$-\frac{\partial A_{z1}}{\partial t} - U_{x1} \frac{\partial A_{z0}}{\partial x} = \eta J_{z1}. \quad (12.20)$$

The current has the form

$$\mu_0 \mathbf{J}_z = \hat{\mathbf{z}} \cdot \nabla \times \mathbf{B} = \hat{\mathbf{z}} \cdot \nabla \times (\nabla \times A_z \hat{\mathbf{z}}) = \hat{\mathbf{z}} \cdot \nabla \times (\nabla A_z \times \hat{\mathbf{z}}) = -\nabla_{\perp}^2 A_z, \quad (12.21)$$

where the subscript $\perp$ means perpendicular to $\hat{\mathbf{z}}$. Thus, the induction equation becomes

$$\frac{\partial A_{z1}}{\partial t} + U_{x1} \frac{\partial A_{z0}}{\partial x} = \frac{\eta}{\mu_0} \nabla_{\perp}^2 A_{z1}. \quad (12.22)$$

To proceed, it is necessary to express the perturbed velocity U_{x1} in terms of A_{z1} ; this relation is obtained from the equation of motion.

While we could just plow ahead and manipulate the equation of motion to obtain U_{x1} in terms of A_{z1} , it is more efficient to exploit the incompressibility relation. In two-dimensional hydrodynamics, incompressibility simplifies flow dynamics so that flow is described by two related scalars, the stream-function f and the vorticity Ω . For two-dimensional motion in the $x-y$ plane of interest here, the general incompressible velocity can be expressed as

$$\mathbf{U} = \nabla f \times \hat{\mathbf{z}}, \quad (12.23)$$

since

$$\nabla \cdot \mathbf{U} = \nabla \cdot (\nabla f \times \hat{\mathbf{z}}) = \hat{\mathbf{z}} \cdot \nabla \times \nabla f = 0. \quad (12.24)$$

The vorticity is the curl of the velocity and because the velocity lies in the $x-y$ plane, the vorticity vector is in the z direction. The vorticity magnitude Ω is given by

$$\Omega = \hat{\mathbf{z}} \cdot \nabla \times \mathbf{U} = \hat{\mathbf{z}} \cdot \nabla \times (\nabla f \times \hat{\mathbf{z}}) = \nabla \cdot [(\nabla f \times \hat{\mathbf{z}}) \times \hat{\mathbf{z}}] = -\nabla_{\perp}^2 f, \quad (12.25)$$

where $\perp$ means perpendicular with respect to z . Given the vorticity, f can be found by solving the Poisson-like Eq. (12.25), and then, knowing f , the velocity can be evaluated using Eq. (12.23). Appropriate boundary conditions must be specified for both f and Ω ; these boundary conditions are that the vorticity is antisymmetric in x and is only large in the vicinity of $x = 0$ as indicated in Fig. 12.2(c).

The curl of the equation of motion provides the vorticity evolution and also annihilates ∇P ; this elimination of P from consideration is why the vorticity/stream-line method is a more efficient approach than direct solution of the equation of motion.

Let us now solve for U_{1x} following this procedure. The linearized equation of motion is

$$\rho_0 \frac{\partial U_1}{\partial t} = (\mathbf{J} \times \mathbf{B})_1 - \nabla P_1; \quad (12.26)$$

taking the curl and dotting with $\hat{z}$ gives

$$\begin{aligned}\rho_0 \frac{\partial \Omega_1}{\partial t} &= \hat{z} \cdot \nabla \times (\mathbf{J} \times \mathbf{B})_1 \\ &= \hat{z} \cdot \nabla \times [J_z \hat{z} \times (\nabla A_z \times \hat{z})]_1 \\ &= \hat{z} \cdot \nabla \times (J_z \nabla A_z)_1 \\ &= \hat{z} \cdot (\nabla J_z \times \nabla A_z)_1.\end{aligned}\quad (12.27)$$

Using Eq. (12.21) this can be written as

$$\frac{\partial \Omega_1}{\partial t} = \frac{1}{\mu_0 \rho_0} \hat{z} \cdot [\nabla A_z \times \nabla (\nabla_{\perp}^2 A_z)]_1. \quad (12.28)$$

From Fig. 12.2(c) it is expected that the vortices have significant amplitude only in the vicinity of where the current bars are deforming and that at large $|x|$ there will be negligible vorticity. Thus, it is assumed that the vorticity evolution equation has the following behavior:

1. Inner (tearing/reconnection) region: here it is assumed that the perturbation has much steeper gradients than the equilibrium so

$$\frac{|\nabla (\nabla_{\perp}^2 A_{z0})|}{|\nabla A_{z0}|} \ll \frac{|\nabla (\nabla_{\perp}^2 A_{z1})|}{|\nabla A_{z1}|}. \quad (12.29)$$

This allows Eq. (12.28) to be approximated as

$$\begin{aligned}\frac{\partial \Omega_1}{\partial t} &\simeq \frac{1}{\mu_0 \rho_0} \hat{z} \cdot [\nabla A_{z0} \times \nabla (\nabla_{\perp}^2 A_{z1})] \\ &= \frac{1}{\mu_0 \rho_0} \frac{dA_{z0}}{dx} \frac{\partial}{\partial y} (\nabla_{\perp}^2 A_{z1}),\end{aligned}\quad (12.30)$$

which shows that J_{z1} crossed with B_{y0} generates vorticity. Since B_{y0} is antisymmetric with respect to x , the vortices have the assumed antisymmetry. Furthermore, because J_{z1} is symmetric with respect to x and localized in the vicinity of $x = 0$ the vortices are localized to the vicinity of $x = 0$.

2. Outer (ideal) region: here $\Omega_1 \simeq 0$ is assumed so Eq. (12.28) becomes

$$\frac{dA_{z0}}{dx} \frac{\partial}{\partial y} (\nabla_{\perp}^2 A_{z1}) - \frac{\partial A_{z1}}{\partial y} \frac{d^3 A_{z0}}{dx^3} = 0, \quad (12.31)$$

which is a specification for A_{z1} in the outer region for a given A_{z0} . Thus, it is effectively assumed that the outer perturbed field is force-free, i.e., $(\mathbf{J} \times \mathbf{B})_1 = 0$. This is consistent with there being no generation of vorticity in the outer region.

The perturbed quantities will now be assumed to have the space-time dependence

$$\begin{aligned} A_{z1} &= A_{z1}(x)e^{iky+\gamma t} \\ \Omega_1 &= \Omega_1(x)e^{iky+\gamma t} \end{aligned} \quad (12.32)$$

so Eq. (12.30) gives the inner region vorticity as

$$\Omega_1 = \frac{1}{\mu_0 \gamma \rho_0} \frac{dA_{z0}}{dx} ik (\nabla_{\perp}^2 A_{z1}) = -\frac{1}{\gamma \rho_0} \frac{dA_{z0}}{dx} ik J_{z1}. \quad (12.33)$$

This satisfies all the geometric conditions noted earlier, namely the antisymmetric dependence on x , the localization near $x = 0$, and, consistent with Fig. 12.2(c), a y -periodicity 90° out of phase with the periodicity of J_{z1} .

Using Eq. (12.23), it is seen that

$$U_{x1} = \frac{\partial f_1}{\partial y} = ik f_1. \quad (12.34)$$

The stream-function f_1 is a solution of the Poisson-like system Eq. (12.25) and Eq. (12.33),

$$\frac{\partial^2 f_1}{\partial x^2} - k^2 f_1 = \frac{1}{\gamma \rho_0} \frac{dA_{z0}}{dx} ik J_{z1}. \quad (12.35)$$

Since the perturbed current peaks at $x = 0$ and has a width of the order of ϵ , it may be characterized by the Gaussian profile

$$J_{z1} \simeq \frac{\lambda}{\epsilon \sqrt{\pi}} e^{-x^2/\epsilon^2}, \quad (12.36)$$

where

$$\lambda = \int_{layer} J_{z1} dx \quad (12.37)$$

is the total perturbed current in the tearing layer. The gradient of the vector potential can be written as

$$\frac{dA_{z0}}{dx} = -B_{y0}(x) \simeq -\frac{x}{L} B'_{y0}. \quad (12.38)$$

Assuming that the tearing layer is very narrow gives

$$\frac{\partial^2 f_1}{\partial x^2} \gg k^2 f_1 \quad (12.39)$$

so Eq. (12.35) becomes

$$\frac{\partial^2 f_1}{\partial x^2} = -\frac{B'_{y0}}{\gamma L \rho_0} ik \frac{\lambda}{\epsilon \sqrt{\pi}} x e^{-x^2/\epsilon^2} = \frac{ik B'_{y0}}{2 \gamma L \rho_0} \frac{\lambda \epsilon}{\sqrt{\pi}} \frac{d}{dx} e^{-x^2/\epsilon^2}. \quad (12.40)$$

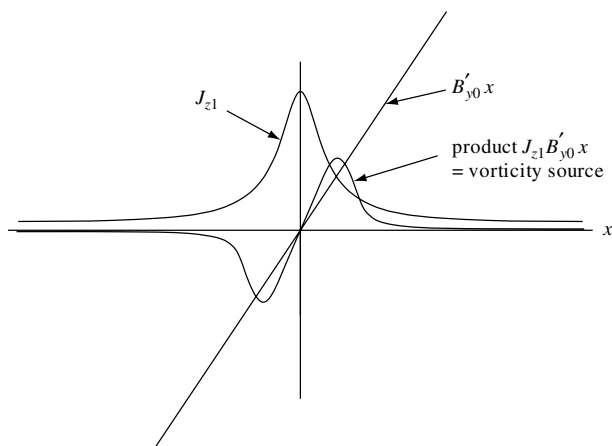


Fig. 12.3 Product of symmetric perturbed current with antisymmetric equilibrium field results in antisymmetric vorticity source localized near $x = 0$.

The profiles of J_{z1} , $B_{y0}(x)$, and their product (right-hand side of Eq. (12.40)) are shown in Fig. 12.3.

Integrating Eq. (12.40) with respect to x gives

$$\frac{\partial f_1}{\partial x} = \frac{ikB'_{y0}}{2\gamma L\rho_0} \frac{\lambda\epsilon}{\sqrt{\pi}} e^{-x^2/\epsilon^2}, \quad (12.41)$$

which incidentally gives $U_{1y} = -\partial f_1/\partial x$. Since it is desired to find the magnitude of U_{1x} in the region $x \sim \epsilon$, a rough “order of magnitude” integration of Eq. (12.41) in this region gives

$$f_1 \sim \frac{ikB'_{y0}\lambda\epsilon^2}{2\gamma L\sqrt{\pi}\rho_0} \text{sign}(x) \text{ for } x \sim \epsilon \quad (12.42)$$

and so

$$U_{x1} \sim -\frac{k^2 B\lambda\epsilon^2}{2\gamma L\sqrt{\pi}\rho_0} \text{sign}(x) \text{ for } x \sim \epsilon. \quad (12.43)$$

The amplitude factor λ can be expressed using Eq. (12.21) as

$$\begin{aligned} \lambda &= -\frac{1}{\mu_0} \int_{\text{layer}} \nabla_{\perp}^2 A_{z1} dx \\ &\simeq -\frac{1}{\mu_0} \int_{\text{layer}} \frac{\partial^2 A_{z1}}{\partial x^2} dx \\ &= -\frac{1}{\mu_0} \left[\left(\frac{\partial A_{z1}}{\partial x} \right)_+ - \left(\frac{\partial A_{z1}}{\partial x} \right)_- \right], \end{aligned} \quad (12.44)$$

where the subscripts $\pm$ mean evaluated at $x = \pm\epsilon$. For purposes of joining to the outer ideal solution, the normalized jump derivative is defined as

$$\Delta' = \frac{\left(\frac{\partial A_{z1}}{\partial x}\right)_+ - \left(\frac{\partial A_{z1}}{\partial x}\right)_-}{A_{z1}(0)} \quad (12.45)$$

so

$$\lambda = -\frac{\Delta'}{\mu_0} A_{z1}(0). \quad (12.46)$$

The velocity becomes

$$U_{x1} \sim \frac{k^2 B \Delta' \epsilon^2}{2\gamma L \sqrt{\pi} \mu_0 \rho_0} A_{z1}(0) \text{sign}(x) \quad (12.47)$$

and using Eq. (12.36), the current density becomes

$$J_{z1} \sim -\frac{\Delta'}{\epsilon \mu_0 \sqrt{\pi}} A_{z1}(0). \quad (12.48)$$

We now repeat the induction equation, Eq. (12.20),

$$-\frac{\partial A_{z1}}{\partial t} - U_{x1} \frac{\partial A_{z0}}{\partial x} = \eta J_{z1} \quad (12.49)$$

and substitute for U_{x1} , J_{z1} and assume that $\partial A_{z0}/\partial x = -B_{y0} \simeq -B\epsilon/L$ in the tearing layer. This gives

$$\underset{\#1}{\gamma} - \underset{\#2}{\frac{k^2 B^2 \Delta' \epsilon^3}{2\gamma L^2 \sqrt{\pi} \mu_0 \rho_0}} = \underset{\#3}{\frac{\eta \Delta'}{\epsilon \mu_0 \sqrt{\pi}}}, \quad (12.50)$$

where the terms have been numbered for reference in the following discussion.

In the ideal plasma limit, terms #1 and #2 balance each other while term #3 is small; this gives the frozen-in condition. At exactly $x = 0$, term #2 vanishes and so terms #1 and #3 must balance each other, resulting in diffusion of the magnetic field. At the edge of the tearing layer, i.e., at the transition from the ideal limit to the diffusive limit, *all three terms are of the same size*. Thus, the three terms may be equated; this gives two equations that may be solved for γ and ϵ with Δ' as a parameter (Furth, Killeen, and Rosenbluth 1963). Equating terms #1 and #3 gives

$$\gamma = \frac{\eta \Delta'}{\epsilon \mu_0 \sqrt{\pi}} \quad (12.51)$$

while equating terms #2 and #3 gives

$$\gamma = \frac{(kB')^2 \epsilon^4}{2\eta\rho_0}, \quad (12.52)$$

where $B' = B/L$ is the derivative of the equilibrium field at $x = 0$. Equating these last two equations to eliminate γ gives the width of the tearing layer to be

$$\epsilon \simeq \left[\frac{2\eta^2 \rho_0 \Delta'}{\mu_0 \sqrt{\pi} (kB')^2} \right]^{1/5}. \quad (12.53)$$

Substituting ϵ back into Eq. (12.51) gives

$$\gamma = 0.55 (\Delta')^{4/5} \left[\frac{\eta}{\mu_0} \right]^{3/5} \left[\frac{(kB')^2}{\rho_0 \mu_0} \right]^{1/5}. \quad (12.54)$$

This result can be put in a more physically intuitive form by defining characteristic times for ideal processes and for resistive processes. The characteristic time for ideal processes is the Alfvén time τ_A , defined as the time to move the characteristic length L when traveling at the Alfvén velocity, i.e.,

$$\tau_A^{-1} = \frac{v_A}{L} = \frac{\sqrt{B^2/\rho_0 \mu_0}}{L} = \sqrt{\frac{(B')^2}{\rho_0 \mu_0}}. \quad (12.55)$$

The Alfvén time is the characteristic time of ideal MHD and is typically a very fast time. The characteristic time for resistive processes τ_R is defined as the time to diffuse resistively a distance L , so

$$\tau_R^{-1} = \frac{\eta}{L^2 \mu_0}. \quad (12.56)$$

For nearly ideal plasmas the resistive time scale is very slow. Using these definitions, Eq. (12.54) can be written as (Furth, Killeen and Rosenbluth 1963)

$$\gamma = 0.55 (\Delta' L)^{4/5} (kL)^{2/5} \tau_R^{-3/5} \tau_A^{-2/5}. \quad (12.57)$$

All that is needed now is Δ' . This jump condition is found from Eq. (12.31) which gives the form of A_{z1} in the ideal region outside the tearing layer. This can be expressed as

$$\nabla_{\perp}^2 A_{z1} + \left[B_{y0}^{-1} \frac{d^2 B_{y0}}{dx^2} \right] A_{z1} = 0, \quad (12.58)$$

which shows that the equilibrium magnetic field acts like a “potential” for the perturbed vector potential “wavefunction.” If boundary conditions are specified at large $|x|$ for the perturbed vector potential, then there will typically be a

discontinuity in the first derivative of A_{z1} at $x = 0$; this discontinuity gives Δ' . The jump depends on the existence of a localized equilibrium current since

$$\frac{d^2 B_{y0}}{dx^2} = \mu_0 \frac{dJ_{z0}}{dx}. \quad (12.59)$$

Equation (12.58) must in general be solved numerically.

The main result, as given by Eq. (12.57), is that if $\Delta' > 0$ an instability develops having a growth rate *intermediate* between the fast Alfvén time scale and the slow resistive time scale. Since a nearly ideal plasma is being considered, η is extremely small. The width of the tearing layer is therefore very narrow, since, as shown by Eq. (12.53), this width is proportional to $\eta^{2/5}$.

12.5 Generalization of tearing to sheared magnetic fields

The sheet current discussed above can occur in real situations but is a special case of the more general situation where the equilibrium magnetic field does not have a null, but instead is simply sheared. This means that the equilibrium magnetic field is straight, has components in both the y and z directions, and has a direction that is a varying function of x . The sheared situation thus has a uniform magnetic field in the z direction and, instead of the current being concentrated in a sheet, there is simply a non-uniform $B_{y0}(x)$.

There is a non-trivial difference between the special field-null equilibria and the more general types of equilibria where there is no field null. This difference occurs because equilibria involve balancing the magnetic and hydrodynamic pressures in such a way that $P + B^2/2\mu_0$ is continuous within the plasma and also across the plasma boundary. If a field null exists, then equilibrium consists of magnetic pressure $B^2/2\mu_0$ exterior to the null balancing hydrodynamic pressure P at the null and so a reconnection region involving a field null must have $\beta \simeq 1$. This situation occurs in the reconnection region associated with the Earth's magnetotail and has also been studied in certain laboratory plasma experiments (Trintchouk *et al.* 2003). However, in the more general case where there is no field null, B^2 is nearly continuous across the reconnection layer. This means that P will be small, in which case $\beta \ll 1$. Thus, the assumption of low β precludes the possibility of a field null and so precludes the possibility of the simple current sheet discussed in the previous section.

The more general situation where the equilibrium magnetic field has the form

$$\mathbf{B}_0 = B_{y0}(x)\hat{y} + B_{z0}\hat{z} \quad (12.60)$$

would thus be appropriate for low- β plasmas such as tokamaks, spheromaks, and the solar corona. A non-trivial feature of this situation is that, unlike the

previously considered sheet current equilibrium, here $B_{y0}(x)$ does not vanish at any particular x . Instead, as will be seen later, what matters is the vanishing of $\mathbf{k} \cdot \mathbf{B}_0$. Equation (12.60) can be used as a slab representation of the straight cylindrical geometry equilibrium field

$$\mathbf{B}_0 = \nabla\psi_0(r) \times \nabla z + B_{z0}\hat{z}. \quad (12.61)$$

Equation (12.61) in turn can be thought of as the straight cylindrical approximation of a toroid with z corresponding to the toroidal angle (see Eq. (9.34)).

Figure 12.4 shows the magnetic field given by Eq. (12.60) as viewed in the $x = 0$ plane. In the sheet current analysis discussed in the previous section, the perturbed vector potential pointed in the z direction and was periodic in the y direction. This corresponded to having $\mathbf{k} \cdot \mathbf{A}_1 = 0$ so that the wavevector was orthogonal to the vector potential and both were orthogonal to x . Other important properties were that $\mathbf{k} \cdot \mathbf{B}$ vanished at the reconnection layer, the fluid vorticity vector was pointed in the z direction, the perturbed currents and perturbed magnetic fields were such that $J_1/J_0 \gg B_1/B_0$ in the reconnection layer, and $J_1/J_0 \sim B_1/B_0$ in the exterior region.

These relationships and approximations are generalized here and, in particular, it is assumed all perturbed quantities have functional dependence $\sim g(x) \exp(ik_y y + ik_z z + \gamma t)$. As before, the vorticity equation is the curl of the linearized equation of motion, i.e.,

$$\rho_0 \frac{\partial \Omega_1}{\partial t} = \nabla \times (\mathbf{J}_1 \times \mathbf{B}_0 + \mathbf{J}_0 \times \mathbf{B}_1) \quad (12.62)$$

and in the reconnection layer where $J_1/J_0 \gg B_1/B_0$ this becomes

$$\begin{aligned} \rho_0 \frac{\partial \Omega_1}{\partial t} &= \nabla \times (\mathbf{J}_1 \times \mathbf{B}_0) \\ &= \mathbf{B}_0 \cdot \nabla \mathbf{J}_1 - \mathbf{J}_1 \cdot \nabla \mathbf{B}_0 \end{aligned} \quad (12.63)$$

since $\nabla \cdot \mathbf{B}_0 = 0$ and $\nabla \cdot \mathbf{J}_1 = 0$.

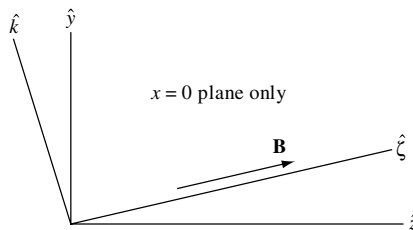


Fig. 12.4 Tilted coordinate system for general sheared field.

An essential feature of the reconnection topology is that the vorticity must be antisymmetric about the reconnection layer, as can be seen from examination of the fluid flow vectors in Fig. 12.2(c). Since the vorticity is created by the torque (i.e., the curl of the force), it is clear that the torque must be antisymmetric about the reconnection layer. As will be seen in the next paragraphs, the condition that the torque is antisymmetric does not imply that either $\mathbf{B}_0$ or $\mathbf{J}_1$ are antisymmetric, but rather implies some more subtle conditions.

Thus, if $x = 0$ is defined to be the location of the reconnection layer, then Ω_1 must be an odd function of x and therefore must vanish at $x = 0$. Hence, the right-hand side of Eq. (12.63) must vanish at $x = 0$ and, since there need be no particular functional relationship between $\mathbf{B}_0$ and its gradient, the two terms on the right-hand side of Eq. (12.63) must separately vanish. Therefore, one of the requirements is to choose the origin of the x axis such that $\mathbf{B}_0 \cdot \nabla \mathbf{J}_1 = 0$ at $x = 0$ or, equivalently,

$$\mathbf{k} \cdot \mathbf{B}_0 = 0 \text{ at } x = 0 \quad (12.64)$$

so that

$$k_y B_{y0}(0) + k_z B_{z0} = 0. \quad (12.65)$$

Having $\mathbf{k} \cdot \mathbf{B}$ vanish at the reconnection layer is physically reasonable, since finite $\mathbf{k} \cdot \mathbf{B}$ implies a periodic bending of the equilibrium field; such a bending absorbs energy and so is stabilizing. Having $\mathbf{k} \cdot \mathbf{B}$ vanish is like letting the instability cleave the system at a weak point so the instability can develop without requiring much free energy.

The second requirement for the right-hand side of Eq. (12.63) to vanish is to have $\mathbf{J}_1 \cdot \nabla \mathbf{B}_0 = 0$. If the flow is incompressible, then the perturbed magnetic field must be orthogonal to the equilibrium magnetic field and since the perturbed current is the curl of the perturbed magnetic field, the perturbed current must be parallel to the equilibrium magnetic field (this is essentially shear Alfvén wave physics). Since $\mathbf{B}_0$ is assumed straight, $\mathbf{B}_0 \cdot \nabla \mathbf{B}_0 = 0$ and, since $\mathbf{J}_1$ is parallel to $\mathbf{B}_0$, it is seen that $\mathbf{J}_1 \cdot \nabla \mathbf{B}_0 = (J_1/B_0) \mathbf{B}_0 \cdot \nabla \mathbf{B}_0 = 0$. Because $\mathbf{J}_1$ is parallel to $\mathbf{B}_0$, $\mathbf{J}_1$ must also be straight and so a Coulomb gauge vector potential will be parallel to $\mathbf{J}_1$, since

$$\begin{aligned} \mu_0 \mathbf{J}_1 &= \nabla \times \nabla \times \mathbf{A}_1 \\ &= \nabla \nabla \cdot \mathbf{A}_1 - \nabla^2 \mathbf{A}_1 \\ &= -\nabla^2 \mathbf{A}_1 \\ &= -\hat{\zeta} \nabla^2 \mathbf{A}_1 \end{aligned} \quad (12.66)$$

can be satisfied by having both $\mathbf{J}_1$ and $\mathbf{A}_1$ in the direction of $\hat{\zeta}$, where $\hat{\zeta} = \mathbf{B}_0(0)/B_0(0)$ does not depend on position.

It is therefore assumed that $\mathbf{A}_1$ is parallel to $\mathbf{B}_0(0)$ and so

$$\mathbf{A}_1(x, y, z, t) = A_{1\zeta}(x) \hat{\zeta} e^{ik_y y + ik_z z + \gamma t}. \quad (12.67)$$

Thus, $\mathbf{k} \cdot \mathbf{A}_1 = 0$ in this tilted coordinate system since $\mathbf{k} \cdot \hat{\zeta} = \mathbf{k} \cdot \mathbf{B}_0(0)/|\mathbf{B}_0(0)|$. Also $\nabla \cdot \mathbf{A}_1 = 0$ so the Coulomb gauge assumption is satisfied.

Reconsideration of the right-hand side of Eq. (12.63) shows that although the two terms both vanish at $x = 0$ there is a difference in the geometrical dependence of these terms. In particular, since $\mathbf{J}_1$ is in the $\hat{\zeta}$ direction, it has no x component, whereas $\mathbf{B}_0$ depends only on x so

$$\mathbf{J}_1 \cdot \nabla \mathbf{B}_0 = \left(J_{y1} \frac{\partial}{\partial y} + J_{z1} \frac{\partial}{\partial z} \right) \mathbf{B}_0(x) = 0. \quad (12.68)$$

Thus, Eq. (12.63) reduces to

$$\begin{aligned} \rho \frac{\partial \Omega_1}{\partial t} &= \mathbf{B}_0 \cdot \nabla \mathbf{J}_1 \\ &= i(\mathbf{k} \cdot \mathbf{B}_0) \mathbf{J}_1, \end{aligned} \quad (12.69)$$

where, by assumption, $\mathbf{k} \cdot \mathbf{B}_0$ vanishes at $x = 0$.

Continuing this discussion of the ramifications of the antisymmetry of $\mathbf{k} \cdot \mathbf{B}_0$ about $x = 0$, we now define an artificial reference magnetic field $\bar{\mathbf{B}}$ that is parallel to the real field at $x = 0$, but has no shear (i.e., has no x dependence). The reference field therefore has the form

$$\bar{\mathbf{B}} = B_{y0}(0) \hat{y} + B_{z0} \hat{z} = B_0(0) \hat{\zeta} \text{ for all values of } x. \quad (12.70)$$

We now define $\mathbf{b}_0(x)$ as the difference between the real field and $\bar{\mathbf{B}}$ so that

$$\mathbf{B}_0(x) = \bar{\mathbf{B}} + \mathbf{b}_0(x). \quad (12.71)$$

Thus $\mathbf{b}_0(x)$ has the same x dependence as the B_{y0} field used for the sheet current instability in the previous section in that (i) it is antisymmetric about x and (ii) it is in the y direction. One way of thinking about this is to realize that $\mathbf{b}_0(x)$ is the component of $\mathbf{B}_0(x)$ that is antisymmetric about $x = 0$. With this definition

$$\mathbf{k} \cdot \mathbf{B}_0 = \mathbf{k} \cdot \mathbf{b}_0(x) = k_y b_0 \quad (12.72)$$

and so Eq. (12.69) becomes

$$\gamma \rho_0 \Omega_{\zeta 1} = i k_y b_0 J_{\zeta 1}. \quad (12.73)$$

This equation provides the essence of the dynamics. It shows how the component of the magnetic field that is antisymmetric about $x = 0$ creates fluid vortices that are antisymmetric with respect to x . In particular, since $\mathbf{b}_0(x)$ is an odd function of x and $J_{\zeta 1}$ is an even function of x , $\Omega_{\zeta 1}$ is an odd function of x .

The complete self-consistent description is obtained by, in addition, taking into account the induction equation, which shows how fluid motion acts to create perturbations of the electromagnetic field.

In the sheet current case the component of Ohm's law in the direction of symmetry of the perturbation was considered, i.e., the component in the z direction. Here, the corresponding symmetry direction for the perturbation is the $\hat{\zeta}$ direction and so the relevant component of Ohm's law is the $\hat{\zeta}$ component,

$$\hat{\zeta} \cdot [\mathbf{E}_1 + \mathbf{U}_1 \times \mathbf{B}_0(x)] = \hat{\zeta} \cdot \eta \mathbf{J}_1, \quad (12.74)$$

which becomes

$$-\frac{\partial A_{\xi 1}}{\partial t} + (\hat{y} \times \hat{\zeta} \cdot \mathbf{U}_1) b_0(x) = \eta J_{\xi 1}. \quad (12.75)$$

Since the vorticity vector lies along $\hat{\zeta}$, the incompressible velocity must be orthogonal to $\hat{\zeta}$ and so has the form

$$\mathbf{U}_1 = \nabla f_1 \times \hat{\zeta}. \quad (12.76)$$

Thus,

$$\hat{y} \times \hat{\zeta} \cdot \mathbf{U}_1 = (\hat{y} \times \hat{\zeta}) \cdot (\nabla f_1 \times \hat{\zeta}) = i k_y f_1. \quad (12.77)$$

Taking into account the $\hat{\zeta}$ direction of the vorticity once again, it is seen that $\Omega_1 = \Omega_{\xi 1} \hat{\zeta}$, where

$$\Omega_{\xi 1} = \hat{\zeta} \cdot \nabla \times \mathbf{U}_1 = -\nabla_{\perp}^2 f_1 \quad (12.78)$$

and now $\perp$ means perpendicular to $\hat{\zeta}$.

Substituting for $\Omega_{\xi 1}$ in Eq. (12.73) using Eq. (12.78) gives

$$\nabla_{\perp}^2 f_1 = -\frac{i k_y b_0(x) J_{\xi 1}}{\gamma \rho_0}; \quad (12.79)$$

this equation provides the essential dynamics of fluid vortices that are antisymmetric with respect to x and driven by the torque associated with the non-conservative $\mathbf{J} \times \mathbf{B}$ force. This is essentially the same as Eq. (12.35) and the rest of the analysis is the same as for the sheet current problem except that now k_y is used instead of k and b'_0 is used instead of B'_{y0} since $b_0(x)$ only differs from $B_{y0}(x)$ by a constant, $b'_0 = B'_{y0}$. Thus, using Eq. (12.54) the growth rate will be

$$\gamma = 0.55 (\Delta')^{4/5} \left[\frac{\eta}{\mu_0} \right]^{3/5} \left[\frac{(k_y B'_{y0})^2}{\rho_0 \mu_0} \right]^{1/5}. \quad (12.80)$$

The global system has to be periodic in both the y and z direction in order for well-defined k_y and k_z to exist. In particular, the physical arrangement and

dimensions of the global system determine the quantized spectra of k_y and k_z and so determine the allowed planes where $\mathbf{k} \cdot \mathbf{B}_0$ can vanish. As suggested earlier, the allowed planes can be considered as “cleavage” planes where the magnetic field can most easily become unglued from the plasma.

Let us express this result in the context of toroidal geometry such as that of a tokamak. This is done by letting B_{z0} correspond to the toroidal field B_ϕ and B_y correspond to B_θ the poloidal field. The Alfvén time is now defined in terms of B_ϕ as

$$\tau_A^{-1} = \frac{B_\phi}{a\sqrt{\rho_0\mu_0}}, \quad (12.81)$$

where a is the minor radius. The safety factor, a measure of the twist, is defined as

$$q = \frac{aB_\phi}{RB_\theta} \quad (12.82)$$

so

$$B_\theta = \frac{aB_\phi}{Rq} \quad (12.83)$$

and

$$B'_{y0} \rightarrow -\frac{aB_\phi}{Rq^2} q'. \quad (12.84)$$

Thus, it is possible to replace $k_y \rightarrow m/a$ and

$$\frac{(k_y B'_{y0})^2}{\rho_0\mu_0} \rightarrow \left(\frac{ma}{Rq^2} q'\right)^2 \frac{1}{\tau_A}. \quad (12.85)$$

At the tearing layer $\mathbf{k} \cdot \mathbf{B} = 0$ or

$$\frac{m}{a} B_\theta + \frac{n}{R} B_\phi = 0 \quad (12.86)$$

so

$$q = -\frac{m}{n} \quad (12.87)$$

and Eq. (12.85) becomes

$$\frac{(k_y B'_{y0})^2}{\rho_0\mu_0} \rightarrow \left(\frac{na}{R} \frac{q'}{q}\right)^2 \frac{1}{\tau_A}. \quad (12.88)$$

Thus, Eq. (12.80) becomes

$$\gamma = 0.55 (\Delta' a)^{4/5} \tau_R^{-3/5} \tau_A^{-2/5} \left(\frac{na^2}{R} \frac{q'}{q}\right)^{2/5}, \quad (12.89)$$

where

$$\tau_R^{-1} = \frac{\eta}{a^2 \mu_0}. \quad (12.90)$$

Equation (12.89) shows that the essential source for the tearing instability is q' . From this point of view, the “free energy” is in the gradient of q and so, as the tearing mode uses up this free energy, the q profile will be flattened. The concept that gradients in q drive reconnection is closely related to the concept that relaxation is driven by gradients in the force-free parameter λ discussed with regards to Eq. (10.140). This is because λ is essentially the axial current I flowing through a flux tube with flux Φ . This can be seen by integrating Eq. (10.140) over the cross-section of a current-carrying flux tube to obtain

$$\frac{\mu_0 I}{\Phi} = \frac{\int \nabla \times \mathbf{B} \cdot d\mathbf{s}}{\int \mathbf{B} \cdot d\mathbf{s}} = \lambda. \quad (12.91)$$

If the torus is approximated as a straight cylinder so B_ϕ is the axial field and $B_\theta = \mu_0 I / 2\pi a$ is the azimuthal field then the axial flux is $\Phi = \pi a^2 B_\phi$ and

$$q = \frac{a B_\phi}{R B_\theta} = \frac{2\pi a^2 B_\phi}{R \mu_0 I} = \frac{2\Phi}{R \mu_0 I}. \quad (12.92)$$

We see that

$$q = \frac{2}{\lambda R} \quad (12.93)$$

so

$$q' = -\frac{2}{\lambda^2 R} \lambda'. \quad (12.94)$$

Thus, q and λ are inversely related and the concept that gradients in λ drive instability is essentially equivalent to the concept that gradients in q drive instability.

12.6 Magnetic islands

The tearing instability changes the topology of the magnetic field and causes the formation of magnetic “islands” (Rutherford 1973, Bateman 1978). The equilibrium magnetic field has the form

$$\mathbf{B} = \bar{\mathbf{B}} + b_0(x) \hat{y} = \bar{\mathbf{B}} + x B'_{y0} \hat{y} = \bar{\mathbf{B}} + \hat{z} \times \nabla A_{z0}(x), \quad (12.95)$$

where

$$A_{z0}(x) = \frac{x^2 B'_{y0}}{2}. \quad (12.96)$$

It is seen that

$$\mathbf{B}_\perp \cdot \nabla A_{z0} = 0, \quad (12.97)$$

where $\mathbf{B}_\perp$ is the component perpendicular to z . Thus, the surfaces $A_z = \text{const.}$ give the projection of the field lines in the plane perpendicular to z ; these projections are the Cartesian geometry equivalents to the poloidal flux surfaces of toroidal geometry.

If it is assumed that $k_z \ll k_y$, so ζ is very nearly parallel to z , then it is seen that the tearing instability adds a perturbation to A_z giving

$$A_z(x, y) = \frac{x^2 B'_{y0}}{2} + A_{z1} \cos k_y y. \quad (12.98)$$

A sketch of a set of surfaces of constant $A_z(x, y)$ is shown in Fig. 12.5. These surfaces consist of (i) closed curves called islands, (ii) a separatrix that passes through the X-point, and (iii) open outer surfaces. On the $x = 0$ line an O-point corresponds to a maximum in $\cos k_y y$ (e.g., $k_y y = 0$) and an X-point corresponds to a minimum (e.g., $k_y y = \pm \pi$). The maximum width w of the separatrix can be calculated by noting that at the X-point (i.e., where $\cos(k_y y) = 1$ and $x = 0$)

$$A_z = 0 + A_{z1} \quad (12.99)$$

while at the point of maximum width on the separatrix (i.e., where $\cos(k_y y) = -1$ and $x = w/2$)

$$A_z = \frac{(w/2)^2 B'_{y0}}{2} - A_{z1}. \quad (12.100)$$

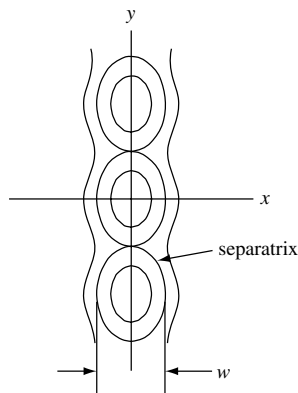


Fig. 12.5 Surfaces of constant A_z showing magnetic islands and separatrix with width w .

Since both these points are on the separatrix, they must have the same value of A_z , and so equating the last two equations gives the island width

$$w = 4 \sqrt{\frac{A_{z1}}{B'_{y0}}} = 4 \sqrt{\left| \frac{B_{x1}}{k_y B'_{y0}} \right|} \quad (12.101)$$

using $B_{x1} = ik_y A_z$. In tokamak terminology we identify $B_{x1} \rightarrow B_{r1}$, where r is the minor radius, and using Eq. (12.84), Eq. (12.87), and $k_y \rightarrow m/a$ this becomes

$$w = 4 \sqrt{\left| \frac{Rq}{nq'} \frac{B_{r1}}{B_\phi} \right|}. \quad (12.102)$$

It is important to realize that the width of the island is much larger than the width of the tearing layer. Since particles tend to be attached to magnetic flux surfaces, the formation of islands means that particles can circulate around the island, thereby causing a flattening of the pressure gradient because the pressure is constant along a magnetic field line.

12.7 Assignments

1. Sweet–Parker type reconnection (Sweet 1958, Parker 1957, Trintchouk, Yamada, Ji, Kulsrud and Carter 2003). Consider the two identical flux-conserving current loops shown in Fig. 12.6(a). Because the system is axisymmetric, the magnetic field can be expressed as

$$\mathbf{B} = \frac{1}{2\pi} \nabla\psi \times \nabla\phi,$$

where ψ is the poloidal flux.

- (a) Explain why there is an attractive force between the current loops (hint: consider the force between parallel currents or between north and south poles of two magnets).
- (b) Define private flux to be a poloidal flux surface that links only one of the current loops (examples are the flux surfaces labeled 1 and 2 in Fig. 12.6(a)). Define public flux to be a flux surface that links both of the current loops (examples are flux surfaces 3, 4, 5 in Fig. 12.6(a)). Define the X-point to be the location in the $z=0$ plane where there is a field null as shown in Fig. 12.6(a); let r_0 be the radius of the X-point. Show by sketching that as the two current loops approach each other in *vacuum*, a private flux surface above the midplane will merge with a private flux surface below to form a public flux surface.
- (c) Show that the flux linked by a circle in the $z=0$ plane with radius r_0 (i.e., the circle follows the locus of the X-point) is the public flux. Argue that if the current loops approach each other, this public flux will increase at the rate that private flux is converted into public flux. By integrating Faraday's law over the surface of

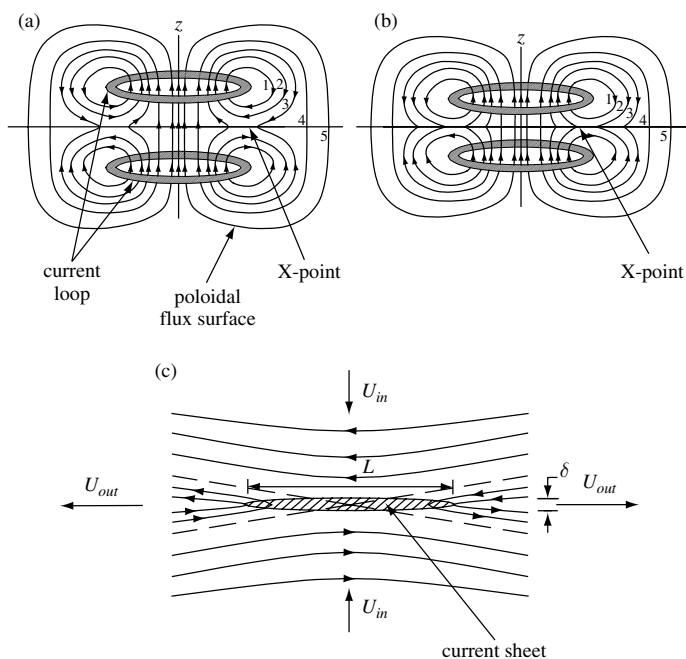


Fig. 12.6 (a) Two identical coaxial current loops with associated poloidal magnetic flux surfaces; (b) current loops approaching each other; (c) detail of X-point region of (b) showing inflows/outflows and current sheet (shaded region) having nominal length L and thickness δ .

this circle in the z -plane with radius r_0 , show there exists a toroidal electric field at the X-point given by

$$E_\phi = -\frac{1}{2\pi r_0} \frac{\partial \psi_{public}}{\partial t},$$

where $\partial \psi_{public} / \partial t$ is the rate of increase of public flux.

- (d) Now suppose that the vacuum is replaced by *perfectly conducting plasma* so that plasma is frozen to flux surfaces. Show from symmetry that the plasma velocity at the X-point must vanish. What is E_ϕ at the X-point if the two current loops are immersed in a plasma that satisfies the ideal Ohm's law,

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = 0?$$

Can there be any conversion of private flux into public flux in an ideal plasma? Sketch contours of private and public flux when two current loops approach each other in an ideal plasma.

- (e) Now suppose the two current loops are immersed in a non-ideal plasma that satisfies the resistive Ohm's law

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}.$$

Can private flux be converted into public flux in this situation, and if so, what is the relationship between the rate of increase of public flux and the resistivity?

- (f) By writing E_ϕ , B_r , B_z , and J_ϕ in terms of the poloidal flux function show that the toroidal component of the resistive Ohm's law can be expressed as

$$\frac{\partial \psi}{\partial t} + \mathbf{U} \cdot \nabla \psi = r^2 \nabla \cdot \left(\frac{\eta}{\mu_0 r^2} \nabla \psi \right).$$

Show that this equation implies that flux convects with the plasma if $\eta = 0$, but diffuses across the plasma if $\mathbf{U} = 0$. Show that the diffusion increases if ψ develops a steep gradient at the location where $\mathbf{U} = 0$. By considering your response to (d) above, discuss how the flux gradient in the z direction might steepen in the vicinity of the X-point. Taking into account the relationship between $\mathbf{J}$ and ψ , show that this corresponds to developing a thin sheet of current in the midplane (current sheet) as shown in Fig. 12.6(c). Show that in order to have an X-point geometry and also satisfy Faraday's law, the flux function must have the form

$$\psi(r, z, t) = \left(1 + \frac{z^2}{2\delta^2} - \frac{(r - r_0)^2}{2L^2} \right) \psi_0 - 2\pi r E_\phi t$$

in the vicinity of the X-point; here ψ_0 is the flux at the X-point at $t = 0$ and r_0 is the radial location of the X-point. What is the relationship between δ and L in vacuum and in a resistive plasma? Is δ larger or smaller than L if the two current loops are approaching each other? If $\delta < L$ is there a current in the vicinity of the X-point? What is the direction of this current with respect to the currents in the two current loops? Will the force on the current loops from the current in the current sheet accelerate or retard the motion of the two current loops towards each other? What is the direction of E_ϕ ? Is ψ increasing or decreasing at the X-point?

- (g) Suppose that the current sheet has a nominal thickness of δ in the z direction and a nominal width of L in the r direction as shown in Fig. 12.6(c). Let U_{in} be the nominal vertical velocity with which the plasma approaches the X-point and let U_{out} be the nominal horizontal velocity with which the plasma leaves the X-point region as shown in Fig. 12.6(c). If the plasma motion is incompressible show that

$$U_{in} L \simeq U_{out} \delta. \quad (12.103)$$

- (h) Consider the transition from the ideal MHD form of Ohm's law to the form at the X-point. Argue that in the region of this transition the terms $\mathbf{U} \times \mathbf{B}$ and $\eta \mathbf{J}$ should have the same order of magnitude (this is essentially the same argument used to analyze Eq. (12.49)). Use this result and Ampère's law to find a relationship between ηJ , B_{in} , and δ . Use this to show

$$U_{in} \sim \eta / \mu_0 \delta \quad (12.104)$$

and explain this result in terms of the convective velocity for motion of flux surfaces outside the current sheet and the "diffusive velocity" for motion of flux surfaces inside the current sheet.

- (i) Using Eq. (9.42) show that the MHD force acting on the plasma is

$$\mathbf{F}_{MHD} = -\frac{\nabla\psi}{(2\pi)^2\mu_0} \nabla \cdot \left(\frac{1}{r^2} \nabla\psi \right).$$

Sketch the direction of this force and indicate where this force is finite (hint: consider where $\mathbf{J}$ is finite). Estimate the work $\int \mathbf{F} \cdot d\mathbf{l}$ done on plasma accelerated through the current sheet by using relationships involving B_{in} , B_{out} , δ , and L . Use this estimate to show that

$$U_{out} \simeq v_{A,in}, \quad (12.105)$$

where $v_{A,in} = B_{in}/\sqrt{\mu_0\rho}$ is the Alfvén velocity at the input. (Hint: let $\Delta\psi$ be the jump in flux experienced by a fluid element as it moves across the current sheet. Note that $B_{out} \sim \Delta\psi/2\pi rL$ and $B_{in} \sim \Delta\psi/2\pi r\delta$, also note that $\nabla \cdot (r^{-2}\nabla\psi)$ can be expressed in terms of B_{in} and δ).

- (j) By eliminating U_{in} and U_{out} between Eqs. (12.103), (12.104), and (12.105) show that

$$\frac{\delta}{L} = \frac{U_{in}}{v_{a,in}} = \frac{1}{\sqrt{S_{in}}},$$

where the inflow region Lundquist number S_{in} is defined as

$$S_{in} = \frac{\mu_0 v_{a,in} L}{\eta}.$$

- (k) If the initial separation between the two current loops is also of the order of L , how much time is required for the two loops to merge, thus completing the reconnection process? Structures in the solar corona have scale lengths $L \sim 10^6$ – 10^8 m and it is observed that Alfvén wave disturbances propagate with velocities $v_A \sim 10^6$ ms⁻¹. The temperature of these structures is 10–100 eV. Assuming a nominal temperature of 50 eV, use Spitzer resistivity to calculate the nominal value of S_{in} . How long would it take for two solar current loops to merge if they are governed by the mechanism discussed here (what is the appropriate time unit to use: seconds, days, weeks, years, ...?). Actual current loops change topology on time scales ranging from minutes to hours – how does the observed time scale compare with the predictions of the resistive reconnection model?